

GEOGRAPHY

INTRODUCTION

Geography is about determining the significance of “place” as it relates to the natural environment, the human environment, and interactions within and between them. To investigate geographic issues, students must analyse the influences and interrelationships that give a place its distinctive characteristics and thus its spatial importance. Geographic analysis also requires an investigation of the economic, environmental, social, and political perspectives that relate to an issue. The application of the concepts of geographic thinking, spatial skills, and the use of field studies are central to the geographic inquiry process and the learning of geography.

Strands

The Grade 11 and 12 geography courses each have five strands. In all of these courses, strand A, Geographic Inquiry and Skill Development, is followed by four content strands, which are organized thematically.

Citizenship Education

The expectations in the Grade 11 and 12 geography courses provide opportunities for students to explore a number of concepts connected to the citizenship education framework (see page 13).

The Concepts of Geographic Thinking

The four concepts of geographic thinking – spatial significance, patterns and trends, interrelationships, and geographic perspective – underpin thinking and learning in all geography courses in the Canadian and world studies program. At least one concept of geographic thinking is identified as the focus for each overall expectation in the content strands of these courses. The following chart describes each concept and provides sample questions related to it. These questions highlight opportunities for students to apply a specific concept in their studies. (See page 16 for a fuller discussion of the concepts of disciplinary thinking.)

Spatial Significance
This concept requires students to determine the importance of a place or region. They explore the connections that exist between the geographical location and physical characteristics of a site and analyse the unique relationships that exist in and between the natural and human environments in a particular place. Students come to understand that the significance of the same place may be different for humans, animals, and plants. Related Questions* – What does a population settlement map tell us about the preferred range of latitude for settlement in Canada? (CGC1P, B2.2) – How does the official plan for your community address urban sprawl? (CGC1D, E2.3) – How might the uniqueness of a destination make it more appealing to some and less appealing to others? (CGG3O, B1.2) – How does the location of a site influence the feasibility of different types of power generation there? (CGR4E, E3.4) – What are some of the challenges that communities in permafrost regions face in providing safe drinking water and disposing of waste? (CGR4M, E2.3)
Patterns and Trends
This concept requires students to recognize characteristics that are similar and that repeat themselves in a natural or human environment (patterns) and characteristics or traits that exhibit a consistent tendency in a particular setting over a period of time (trends). The characteristics may be spatial, social, economic, physical, or environmental. Students analyse connections between characteristics to determine patterns; they analyse connections between those characteristics over time to determine trends. Related Questions – What trends do you see in the use of alternative energy in Ontario? (CGC1P, A1.5) – What pattern or patterns do you see in the location of First Nations reserves across Canada? (CGC1D, D3.1) – What factors have led to population movement in the region? What specific locations have attracted migrants? Why? (CGD3M, E2.1) – In what ways has technological change amplified the impact of population growth on the natural environment? In what ways has it diminished it? (CGW4U, C2.2)

* These questions are drawn directly from the overview charts that precede the geography courses and from the sample questions that accompany many specific expectations. To highlight the continuity between the geography courses in Grade 11 and 12 and those in Grade 9, and to show possible development in the use of the concepts of geographic thinking over those grades, the chart includes some questions from the Grade 9 geography curriculum as well.

Interrelationships

This concept requires students to explore connections within and between natural and human environments. The interconnected parts of an environment or environments work together to form a system. Students must understand the relationships that exist within a system and then critically analyse the relationships between systems in order to determine the impact they have on one another.

Related Questions

- How does surrounding farmland support a community, and what stresses might the community place on the farmland? (CGC1P, E2.1)
- How might the breaking up of continental ice in Greenland and the Antarctic affect Canada's coastline? (CGC1D, B2.1)
- Why do we need to understand the earth's physical processes in order to predict their potential hazards? (CGF3M, Overview)
- How can international trade policies and corporate actions influence disparities within or between ecumenes? (CGU4M, D1.2)

Geographic Perspective

This concept requires students to consider the environmental, economic, political, and/or social implications of the issues, events, developments, and/or phenomena that they are analysing. In order to solve problems, make decisions or judgements, or formulate plans of action effectively, students need to develop their ability to examine issues from multiple perspectives.

Related Questions

- In what ways can cultural diversity enrich the life of a community? (CGC1P, D1.2)
- Does the financial benefit of extracting natural resources justify related social and/or environmental impacts? (CGC1D, A1.6)
- How might spatial technologies help governments assess the economic, environmental, infrastructural, and social risks of a potential disaster? Why is it important for governments to know and understand these risks? (CGT3O, D1.2)
- Why might some people believe that there is an ethical imperative to share the world's resources equitably? (CGW4C, Overview)
- Why might military and/or government satellite information be shared with or withheld from certain countries or military alliances? What are the implications of doing so? (CGO4M, D2.1)

The Geographic Inquiry Process

In each of the geography courses in the Canadian and world studies curriculum, strand A focuses explicitly on the geographic inquiry process, guiding students in their investigations of issues, events, developments, and/or various geographic phenomena. This process is *not* intended to be applied in a linear manner: students will use the applicable components of the process in the order most appropriate for them and for the task at hand. Although strand A covers all of the components of the inquiry process, it is important to note that students apply skills associated with the inquiry process throughout the content strands in each course. (See page 32 for a fuller discussion of the inquiry process in the Canadian and world studies program.)

The following chart identifies ways in which students may approach each of the components of the geographic inquiry process.

Formulate Questions
Students formulate questions: – to explore various events, developments, issues, and/or phenomena that are related to the overall expectations in order to identify the focus of their inquiry – to help them determine which key concept or concepts of geographic thinking are relevant to their inquiry – that reflect the selected concept(s) of geographic thinking – to develop criteria that they will use in evaluating data, evidence, and/or information; in making judgements, decisions, or predictions; in reaching conclusions; and/or in solving problems
Gather and Organize
Students: – collect relevant qualitative and quantitative data, evidence, and information from field studies^a and a variety of primary and secondary sources,^b including visuals^c and community resources^d – determine if their sources are credible, accurate, and reliable – identify the purpose and intent of each source – identify the points of view in the sources they have gathered – use a variety of methods to organize the data, evidence, and/or information they have gathered – record the sources of the data, evidence, and information they are using – decide whether they have collected enough data, evidence, and/or information for their inquiry

a. Field studies may include, but are not limited to, studies in local neighbourhoods, school grounds, and various sites that allow students to explore different land uses (e.g., recreational, commercial, industrial, and transportation uses).

b. Primary sources may include, but are not limited to, census data, land claims, letters, photographs, speeches, and works of art. Secondary sources may include, but are not limited to, documentaries and other films, news articles, reference books, and most websites.

c. Visuals may include, but are not limited to, satellite images, maps, globes, models, graphs, and diagrams.

d. Community resources may include, but are not limited to, local conservation areas, resources from community groups and associations, government resources, and local plans.

Interpret and Analyse
Students: – analyse data, evidence, and information, applying the relevant concepts of geographic thinking (see preceding chart) – use different types of graphic organizers to help them interpret and/or analyse their data, evidence, and information – identify the key points or ideas in each source – analyse graphs, charts, diagrams, and maps – construct graphs, charts, diagrams, and maps to help them analyse the issue, event, development, or phenomenon they are investigating – analyse their sources to determine the importance of an issue, event, development, or phenomenon for individuals or groups – identify biases in individual sources – determine if all points of view are represented in the source materials as a whole, and which, if any, are missing
Evaluate and Draw Conclusions
Students: – synthesize data, evidence, and information, and make informed, critical judgements based on that data, evidence, and information – determine the short- and long-term impact of an event, development, issue, or phenomenon on people and/or places – reach conclusions about the subject of their inquiry, and support them with their data, evidence, and information – make predictions based on their data, evidence, and information – determine the ethical implications of an issue or action – determine the action required, where appropriate
Communicate
Students: – use appropriate forms (e.g., oral, visual, written, multimedia) for different audiences and purposes – communicate their arguments, conclusions, predictions, and plans of action clearly and logically – use geographical terminology and concepts correctly and effectively – cite sources, using appropriate forms of documentation

World Geography: Urban Patterns and Population Issues, Grade 12

University/College Preparation

CGU4M

The world's population is growing, it is moving and intermixing, and it is increasingly found in cities. This course explores these changes and the challenges that come with them. It investigates the forces that are shaping the world's communities, the patterns of interaction between them, the quality of life within them, and their impact on the world around them. Students will apply the concepts of geographic thinking, the geographic inquiry process, and spatial skills and technologies as they investigate issues related to population change and urban life and propose ways of enhancing the sustainability of communities around the world.

Prerequisite: Any university, university/college, or college preparation course in Canadian and world studies, English, or social sciences and humanities

OVERVIEW

The course has five strands. Instruction and learning related to the expectations in strand A are to be interwoven with instruction and learning related to expectations from the other four strands. Strand A must not be seen as independent of the other strands. Student achievement of the expectations in strand A is to be assessed and evaluated *throughout* the course.

Strand A

A. Geographic Inquiry and Skill Development
Overall Expectations
A1. Geographic Inquiry: use the geographic inquiry process and the concepts of geographic thinking when investigating issues affecting ecumenes
A2. Developing Transferable Skills: apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify careers in which a background in geography would be an asset

(continued)

Overview *(continued)*

Throughout this course, when planning instruction, teachers should weave the expectations from strand A in with the expectations from strands B–E.

Strands B–E

Overall Expectations and Related Concepts of Geographic Thinking	Big Ideas*	Framing Questions*
B. Spatial Organization of Ecumenes		
B1. Increasing Liveability: assess the liveability of ecumenes in Canada and other parts of the world, and analyse factors that influence liveability (FOCUS ON: <i>Spatial Significance; Patterns and Trends</i>)	We need to support and maintain the development of liveable communities.	What makes a community liveable? How might individuals' criteria for liveability vary depending on their values, beliefs, and ideas?
B2. Processes that Shape Ecumenes: describe interrelationships between the environmental, social, economic, and political processes that shape ecumenes (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	Many local, national, and international factors have an impact on the growth and development of communities.	Why might some people and groups not support development within a community?
B3. Understanding Ecumenes: describe the spatial distribution of land use in human settlements according to selected theories (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	Land-use patterns and trends can help us understand interrelationships within and between human settlements.	Is land-use distribution and classification similar throughout the world?
C. Sustainability and Stewardship		
C1. Working towards Sustainability: analyse impacts of human activity and human settlements on the environment, and assess the effectiveness of solutions to these impacts in selected ecumenes (FOCUS ON: <i>Spatial Significance; Geographic Perspective</i>)	Human activity and human settlement have social, environmental, political, and economic consequences.	How do population changes affect the impact of communities on the natural environment?
C2. Stewardship: assess ways in which stewardship practices can contribute to the sustainability of human settlements (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	Individuals, governments, and companies must work together to manage a community's growth sustainably.	What are some challenges associated with balancing human wants, needs, and activities with the sustainability of human settlements? How have some groups addressed these challenges?
C3. Modifying the Environment: describe ways in which human societies modify their local environments in order to meet economic, social, political, and other needs, and assess the effects of these modifications on sustainability (FOCUS ON: <i>Spatial Significance; Geographic Perspective</i>)	Environmental modification and sustainability are potentially conflicting ideas.	Is it possible to modify the environment and maintain sustainability at the same time?

* See page 17 for a discussion of the purpose of big ideas and framing questions.

Overall Expectations and Related Concepts of Geographic Thinking	Big Ideas*	Framing Questions*
D. Systems: Interdependence of Ecumenes		
D1. Policies and Change: analyse impacts of public opinion and policy on interactions within and between ecumenes (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	It is important to understand differing points of view when looking at issues involving communities.	Why is it hard to please everyone when making a community decision? How are global population trends affecting human settlement patterns? What infrastructure needs to be developed to accommodate these population trends?
D2. Population Change and Migration: analyse impacts of population change and migration on the social characteristics and built environment of cities (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	Demographic trends are having a wide range of impacts on cities around the world.	How do various international issues affect the quality of life in communities around the world? Why are some communities not able to address these challenges themselves?
D3. Challenges of Interdependence: analyse the effects of international assistance and formal and informal international economic activity on quality of life in developed and developing countries (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	Countries need to continue to work together to develop plans to address quality of life challenges.	
E. Changing Ecumenes		
E1. Quality of Urban Life: analyse interrelationships between urban environments and quality of life (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	The quality of life in a city can vary greatly from one region to another.	What are some quality of life issues that are specific to large urban centres?
E2. Analysis of Solutions: assess from a geographic perspective the effectiveness of solutions to issues affecting the built environment (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	The effectiveness of solutions to urban problems depends on many factors.	What accounts for the variability of quality of life within a city? Why might a megaproject cause new problems while solving an existing problem?

A. GEOGRAPHIC INQUIRY AND SKILL DEVELOPMENT

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1. Geographic Inquiry:** use the geographic inquiry process and the concepts of geographic thinking when investigating issues affecting ecumenes;
- A2. Developing Transferable Skills:** apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset.

SPECIFIC EXPECTATIONS

A1. Geographic Inquiry

Throughout this course, students will:

- A1.1** formulate different types of questions to guide investigations into issues affecting ecumenes (e.g., factual questions: *How many cities in China have populations greater than 10 million?*; comparative questions: *Which region is experiencing the greatest rate of urbanization?*; causal questions: *How do local economies benefit from seasonal migration?*)
- A1.2** select and organize relevant data and information on geographic issues from a variety of primary and secondary sources (e.g., primary: *raw data from fieldwork, both quantitative and qualitative; statistics; photographs; satellite images*; secondary: *newspapers, books, atlases, geographic magazines, websites, graphs, charts, digital and print maps*), ensuring that their sources represent a diverse range of perspectives

Sample questions: “What are some quality of life indicators that you might use to analyse poverty distribution patterns in different ecumenes? Where might you find such data and information?” “What kinds of data and information do you need to collect in order to assess the impact of natural resource extraction or harvesting on a specific ecumene?” “What types of maps and graphs will help you analyse migration patterns?”

- A1.3** assess the credibility of sources and information relevant to their investigations (e.g., *by considering how the data are constructed to support the author’s point of view, the possible bias of the author, the expertise of the author, the accuracy of the text and supporting data, the intended audience, the purpose of the messaging, the context in which the information was presented*)

Sample questions: “Whose point of view does this source represent? Is the source biased? Have you consulted other sources that represent other points of view? Which source is most credible and why?” “What do the author’s credentials suggest about his or her expertise in the subject?” “Are the sources cited in the references and bibliography reputable and credible?” “Are there any ideas that need cross-checking?” “Do you see any statements that are not supported?”

- A1.4** interpret and analyse data and information relevant to their investigations, using various tools, strategies, and approaches appropriate for geographic inquiry (e.g., *interpret diagrams illustrating theories of urban land use; analyse graphs and charts of quality of life indicators for selected ecumenes to identify trends and correlations; use decision-making templates to analyse points of view on an issue; use graphic organizers to outline various perspectives on immigration policies*)

Sample questions: “What type of graphic organizer would you use to help analyse the impact of rural-to-urban migration in selected countries?” “What data layers and types of

information might you include in a geographic information system (GIS) query in order to analyse inter- and intra-urban transportation networks?" "Is there a correlation between the two variables you plotted? How close is it? Does the correlation imply that changes in one variable caused changes in the other? Is it possible that changes in both variables are caused by other factors? How do you explain data points that are not near the line of best fit? What additional evidence do you need to support a causal connection?" "Is the amount of data sampled in the study enough to produce meaningful results?"

A1.5 use the concepts of geographic thinking (i.e., spatial significance, patterns and trends, interrelationships, geographic perspective) when analysing and evaluating data and information and formulating conclusions and/or judgements about issues affecting ecumenes (e.g., *use the concept of spatial significance to analyse the distribution of urban and rural populations; use the concept of patterns and trends to analyse short- and long-term population trends and phenomena, such as the shift of population to urban centres; use the concept of interrelationships to assess the contributions of various natural and human factors to the formation of a megalopolis; use the concept of geographic perspective to analyse the social, political, economic, and environmental impacts of the arrival of a new industry on a selected ecumene*)

Sample questions: "How might the concept of spatial significance help you determine the best location for a new business and the population threshold needed to support it?" "How might an understanding of patterns and trends help you analyse the impact of a megaproject on a selected region of settlement?" "How might an understanding of interrelationships guide your analysis of the connections between two or more areas of population settlement?" "How can geographic perspective help you analyse the impacts of globalization on a selected ecumene?"

A1.6 evaluate and synthesize their findings to formulate conclusions and/or make informed judgements or predictions about the issues they are investigating

Sample questions: "What did you find out about the relationship between government policy and refugee settlement programs? What conclusions can you draw about why this relationship exists?"

A1.7 communicate their ideas, arguments, and conclusions using various formats and styles, as appropriate for the audience and purpose (e.g., *a debate for classmates on the relocation of Aboriginal communities in Canada, a slide presentation for a Grade 9 class showing the differences between "slums of hope" and "slums of despair", a webcast or podcast for the general public on strategies for improving the liveability of a particular community*)

Sample questions: "What does your audience know about the topic? How much and what kind of information do they need? What format and approach would be most effective in conveying your information to this particular audience?" "What type of graph or map projection conveys the information and intended message most accurately and clearly?" "Are there certain data layers that you can add to a map that would make it communicate your ideas more effectively?"

A1.8 use accepted forms of documentation (e.g., *footnotes, author/date citations, reference lists, bibliographies, annotated bibliographies, credits*) to reference different types of sources (e.g., *websites, blogs, books, articles, films, data*)

A1.9 use appropriate terminology when communicating the results of their investigations (e.g., *vocabulary specific to their inquiry; terminology related to geography and to the concepts of geographic thinking*)

A2. Developing Transferable Skills

Throughout this course, students will:

A2.1 describe ways in which geographic investigation can help them develop skills, including spatial skills and the essential skills in the Ontario Skills Passport (e.g., *reading graphic texts, writing, graphing, computer use, use of spatial technologies, oral communication, numeracy, decision making, planning, management, finding information, problem solving*), that can be transferred to postsecondary opportunities, the world of work, and everyday life

A2.2 apply in everyday contexts skills and work habits developed through geographic investigation (e.g., *ask questions to deepen their understanding of an issue; listen to and consider multiple perspectives when discussing an issue; use spatial skills to determine the best location for a home or business or the best route for a holiday trip; apply work habits such as collaboration to*

share information effectively and determine criteria that need to be considered when making a decision; use organizational skills and work habits to help them establish priorities and manage their time both in class and while doing work in other contexts)

A2.3 apply the concepts of geographic thinking when analysing current events involving geographic issues (e.g., *use the concept of spatial significance to analyse possible reasons for the growth of an urban area; use the concept of patterns and trends to analyse the costs and benefits of major international political or sporting events; use the concept of interrelationships to analyse the connection between migration rates and levels of employment for different population settlements;*

use the concept of geographic perspective to analyse the potential consequences of building an energy megaproject in a selected region of population settlement) in order to enhance their understanding of these issues and their role as informed citizens

A2.4 identify some careers in which a geography background might be an asset (e.g., *community settlement worker, diversity officer, entrepreneur, financial analyst, import business owner, economic development officer, landscape architect, politician, project manager, land surveyor, international aid worker, marketing analyst, urban planner, GIS technician)*

B. SPATIAL ORGANIZATION OF ECUMENES

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1. Increasing Liveability:** assess the liveability of ecumenes in Canada and other parts of the world, and analyse factors that influence liveability (**FOCUS ON:** *Spatial Significance; Patterns and Trends*)
- B2. Processes that Shape Ecumenes:** describe interrelationships between the environmental, social, economic, and political processes that shape ecumenes (**FOCUS ON:** *Spatial Significance; Interrelationships*)
- B3. Understanding Ecumenes:** describe the spatial distribution of land use in human settlements according to selected theories (**FOCUS ON:** *Patterns and Trends; Interrelationships*)

SPECIFIC EXPECTATIONS

B1. Increasing Liveability

FOCUS ON: *Spatial Significance; Patterns and Trends*

By the end of this course, students will:

- B1.1** use a variety of criteria (*e.g., rate of urbanization, cultural and economic characteristics, demographics, transportation, infrastructure*) to assess and compare the liveability of selected ecumenes from around the world

Sample questions: “What characteristics do megalopolises, cities, towns, and villages have in common? How are they different, and how do these differences affect their liveability?” “How do cultural differences and related value systems affect people’s opinions of what constitutes liveability?” “Which criteria are the most useful for assessing the liveability of an ecumene?” “How does the liveability of your community compare with that of a community of similar size in France, Brazil, or Indonesia?” “Why does the liveability of a city depend on one’s personal perspective?”

- B1.2** describe changes in the liveability of Canadian and selected ecumenes in other countries from 1800 to the present, and analyse reasons for these changes (*e.g., economic growth, improvements in access to health care and education, safer drinking water, improved wastewater and transportation infrastructure*)

Sample questions: “What role has technology played in changes to infrastructure? What modern services have required new infrastructure?” “How does infrastructure affect the

liveability of an ecumene?” “Have other places in the world enjoyed improvements in liveability similar to those in most of Canada? Which ones have not? Why might different countries have different criteria for liveability based on their values, priorities, and political and economic trends over time?” “What places in Canada have not enjoyed the same improvements in liveability as the majority of Canadian ecumenes?” “What, in your opinion, are the three most important things that have improved the liveability of Canadian ecumenes? Have the same factors been equally important in improving the liveability of ecumenes in the other countries you have looked at, or have other factors been more important?”

Using spatial skills: Students can support their investigations of changes in liveability over time by layering maps of a city or other ecumene from different time periods and annotating them with information about population and the infrastructure that was available in each time period. They can then make inferences about how existing infrastructure affected liveability and how changes in population and infrastructure led to improvements or declines in liveability.

- B1.3** analyse the liveability of their local community and various strategies for enhancing it (*e.g., live/work land use, improved transit, New Urbanism*)

Sample questions: “Should there be more green space in your community?” “Are recreation facilities adequate?” “How should land uses

in the community be changed to improve liveability?" "Should new developments make it easier for people to walk to their destinations or to drive?" "What is New Urbanism, and what approaches does it use to enhance liveability?" "What are the financial and economic benefits of improving transportation routes and reducing traffic congestion in a large urban centre?"

Using spatial skills: Students can integrate concepts related to this expectation by developing an "official" plan for their community that incorporates specific strategies for enhancing liveability.

B2. Processes that Shape Ecumenes

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

B2.1 explain how the physical characteristics of different sites (*e.g., landforms, climate, soils and vegetation*) encourage or constrain urban growth

Sample questions: "How has settlement in Egypt been influenced by the Nile River?" "How does the location of Las Vegas determine or constrict urban growth in that city?" "How do landforms influence the spatial distribution of human settlements in Japan?" "How does the Fraser River influence urban development in the Greater Vancouver area?"

B2.2 analyse the influence of social factors (*e.g., class, religion, ethnicity, cultural activity*) on the characteristics of a selected ecumene (*e.g., land use, economic activity*)

Sample questions: "What effects do religious practices have on settlement patterns in Jerusalem? What evidence is there of commercial activity to meet the cultural dietary requirements (*i.e., for halal and kosher foods*) of specific groups in the city?" "What impacts does a university have on the surrounding community?" "What causes communities to evolve in unique ways? Why might ways of life sometimes be different in communities that have developed in similar physical environments – communities in the Gobi, Mojave, or Sahara deserts, for example, or mountain communities in Bolivia and Tibet, or equatorial communities in Uganda and Singapore?" "How can political decisions alter the development of a particular ecumene?"

Using spatial skills: Students can support an analysis of the social impacts of immigration on their community or region by annotating a map

of the area with information indicating areas of immigrant settlement and related changes in land use.

B2.3 analyse changes in the economic and political characteristics of communities as they evolve through the urban hierarchy (*e.g., from village to town, city, metropolitan area, megacity, and megalopolis*)

Sample questions: "How do the locations of businesses and industries change as a community evolves through the urban hierarchy from town to city?" "How do laws and by-laws encourage or discourage new businesses?" "How do threshold populations influence economic and political opportunities and needs as a community evolves through the urban hierarchy?" "How does the availability of low-, middle-, and high-order goods within a community relate to the community's position within the urban hierarchy?"

Using spatial skills: Students can calculate the population threshold needed to make it profitable to sell a certain good or service in their community. They can then use the threshold value to determine whether the community has room for more outlets for goods or services of that kind or whether it already has too many.

B3. Understanding Ecumenes

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

B3.1 identify different types of settlements, and describe the criteria used to define them in Canada and other parts of the world

Sample questions: "Is there a difference between the Japanese definition of a city and the Canadian definition?" "How is a town different from a city, other than being smaller?" "How is a megacity different from a megalopolis?"

B3.2 use a variety of maps to compare land-use patterns in cities in developed countries with those in developing countries

Sample questions: "What similarities and differences are evident between the cities you have been investigating? What historical, political, cultural, and economic factors have contributed and/or are contributing to these differences and similarities?" "What does the information you have tell you about the problems

that people in these locations are likely to experience? About their quality of life?"

Using spatial skills: Students can use field study data that they have collected to construct a land-use map of their community. The map can be used as a basis for comparing land uses in their community with those in other cities in the world.

- B3.3** analyse interrelationships between the dominant function of ecumenes (*e.g., government capital, resource centre, manufacturing centre, transportation node, tourist destination*) and their location and land-use patterns, in both developed and developing countries

Sample questions: "What land-use pattern is common to cities with a considerable number of manufacturing industries?" "Do resource-based towns in developed and developing countries have more similarities than differences?" "Why are specialized industries often located together in the same community?" "What characteristics are typical of single-industry towns?"

- B3.4** analyse the spatial distribution of urban hierarchies in selected regions of the world (*e.g., midwestern United States, the Greater Toronto Area, Paris and the surrounding area, Tokyo or Nagasaki and surrounding areas*)

Sample questions: "Are there anomalies that challenge the distribution pattern described in Christaller's central place theory, in which

smaller settlements are clustered around a larger central place in a hexagonal pattern? In what regions of the world would you find examples that don't conform to Christaller's theory?"

Using spatial skills: Students can apply Christaller's central place theory to a selected region to identify different sizes of settlements in a region and analyse their hierarchical and functional relationship to each other. In the analysis, students should consider factors that might challenge the hexagonal shape theory.

- B3.5** explain selected theories and models of urban structure (*e.g., concentric zone, sector, multiple nuclei, irregular pattern*), and evaluate their ability to explain the structure of selected major cities in different parts of the world

Sample questions: "How do different theories or models reflect the time when they were developed?" "Which of these theories works best to explain land use in your community?" "How does the physical environment alter the applicability of the theory or model?" "How does technology alter the way we use space in our community?" "How do we know that a new theory or model is needed or that an existing theory or model needs to be modified?" "What criteria could be used to evaluate a new model or theory of urban land use?"

Using spatial skills: Students can make diagrams to support their analyses of models and theories. Annotations can be added to maps of various urban settlements to illustrate the application of different models or theories.

C. SUSTAINABILITY AND STEWARDSHIP

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1. Working towards Sustainability:** analyse impacts of human activity and human settlements on the environment, and assess the effectiveness of solutions to these impacts in selected ecumenes (**FOCUS ON:** *Spatial Significance; Geographic Perspective*)
- C2. Stewardship:** assess ways in which stewardship practices can contribute to the sustainability of human settlements (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- C3. Modifying the Environment:** describe ways in which human societies modify their local environments in order to meet economic, social, political, and other needs, and assess the effects of these modifications on sustainability (**FOCUS ON:** *Spatial Significance; Geographic Perspective*)

SPECIFIC EXPECTATIONS

C1. Working towards Sustainability

FOCUS ON: *Spatial Significance; Geographic Perspective*

By the end of this course, students will:

- C1.1** analyse environmental impacts of large-scale migration on selected settlements

Sample questions: “How does a large influx of migrants affect resource consumption in a community? What are the effects on waste management and on water and air quality?” “Why do these environmental issues have the potential to become social, political, and economic issues?” “How does the growth of informal settlements such as shantytowns, favelas, and barrios affect the local environment?”

- C1.2** assess the effectiveness of various solutions to environmental problems caused by human activities that affect areas of settlement

Sample questions: “What strategies have been employed to cope with increased volumes of waste? What are their advantages and disadvantages?” “Why would communities either import or export waste?” “What methods are currently used or under consideration for storing nuclear waste?” “Which option would pose less environmental risk for transporting oil through a specified area of settlement: pipelines or rail tank cars?” “How does urban and agricultural runoff affect receiving water bodies in

different parts of the world, and what are some of the steps that governments have taken to control the impacts?”

- C1.3** assess the effects of an ecumene’s natural characteristics on the feasibility of various options for reducing the human impact on the environment (*e.g., number of days of bright sunshine and feasibility of solar power, proximity to volcanoes and hot springs and feasibility of geothermal power*)

C2. Stewardship

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- C2.1** analyse the roles and responsibilities of individuals, corporations, and governments in ensuring the sustainability of communities

Sample questions: “What examples of government commitment to social well-being are evident in your community?” “What incentives do you know about that work to increase individual participation in green waste management programs?” “What social, political, economic, or environmental arguments are used to support or oppose the privatization of water distribution in selected communities?” “Should a corporation be willing to incur costs related to environmental protection that put it at a disadvantage to its competitors? Do governments have a responsibility to create a level playing

field with respect to the environmental obligations of businesses?" "Should governments impose environmental taxes? Should we, as individuals, be willing to pay higher taxes to support sustainable communities?" "How much environmental risk are we willing to accept in order to create or maintain jobs?"

C2.2 evaluate, through research, the effectiveness of selected greening initiatives in urban communities (*e.g., rooftop gardens, community gardens, bike lanes, public transit improvements, alternative energy projects*), and assess the potential for implementing such initiatives in their own community

Sample questions: "What criteria can be used to measure the success of these initiatives?" "How were communities persuaded to adopt these initiatives? What obstacles or barriers did these initiatives face?" "Could any of these initiatives be implemented in our own community? What would the benefits be? Would any members of the community be opposed? Why? How would the initiative be funded?" "How might a community garden unite a neighbourhood?"

Using spatial skills: Students can construct an annotated map of green initiatives in their community, explaining why projects are located where they are and noting any barriers that may inhibit the success of these projects.

C2.3 describe actions that individuals can take to contribute to the sustainability of their own communities

Sample questions: "How is community sustainability linked to personal behaviour? What can you do to increase awareness of these links in your community?" "What can you and other individuals do to reduce your personal contributions to waste generation, energy consumption, and water and air pollution?" "How does buying locally contribute to the social, economic, and environmental sustainability of your community?" "What could you and your fellow students do to make our school 'greener'?"

C3. Modifying the Environment

FOCUS ON: *Spatial Significance; Geographic Perspective*

By the end of this course, students will:

C3.1 describe how populations in different regions have modified their physical environment to improve economic productivity, and assess the impacts of these modifications on the economic,

social, and environmental sustainability of these regions (*e.g., terrace farming in China, reclaimed land in Toronto, diverted rivers in India, polders in the Netherlands*)

Sample questions: "How does the physical environment limit the opportunities of this region? How could the physical environment be modified to reduce these limitations? How would these modifications affect the sustainability of the region? What kinds of compromises would be necessary to achieve the region's economic goals while ensuring its social and environmental sustainability?"

C3.2 describe how populations in different parts of the world have modified their built and natural environments to make them more resilient to impacts from natural disasters (*e.g., hurricanes, earthquakes, floods, volcanoes*)

Sample questions: "What kinds of modifications are commonly made to the built and natural environments in areas that regularly experience hurricanes and flooding?" "How have structures and natural features along the Yangtze River been modified to control flooding?" "Why are people in economically poorer settlement areas more likely to be injured or killed, even in moderate earthquakes, than people in wealthier areas?" "What kinds of modifications will be needed in the future to adapt to climate change?"

C3.3 describe how different cities/regions have modified their built and physical environments in order to host a global event (*e.g., FIFA World Cup, Olympics, world fairs, Pan Am Games*), and assess the impacts of these modifications on the economic, social, and environmental sustainability of these cities/regions

Sample questions: "What immediate impacts did the upgrading of British Columbia's Sea-to-Sky Highway for the 2010 Winter Olympics and the building of the highway to Sochi for the 2014 Olympics have on communities in the region and on the environment? What benefits have these upgrades had since, and were they worth the cost?" "What use is now being made of facilities constructed for Expo 67 in Montreal and the 2008 Olympic Games in Beijing?" "Do such events contribute to the long-term economic, social, and environmental sustainability of a region, or are there better ways of achieving sustainability objectives?"

D. SYSTEMS: INTERDEPENDENCE OF ECUMENES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1. Policies and Change:** analyse impacts of public opinion and policy on interactions within and between ecumenes (**FOCUS ON:** *Patterns and Trends; Interrelationships*)
- D2. Population Change and Migration:** analyse impacts of population change and migration on the social characteristics and built environment of cities (**FOCUS ON:** *Patterns and Trends; Interrelationships*)
- D3. Challenges of Interdependence:** analyse the effects of international assistance and formal and informal international economic activity on quality of life in developed and developing countries (**FOCUS ON:** *Interrelationships; Geographic Perspective*)

SPECIFIC EXPECTATIONS

D1. Policies and Change

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

- D1.1** analyse the influence of different points of view and self-interest on conflicts over local urban issues
Sample questions: “What does NIMBY mean? Should the interests of a neighbourhood take precedence over those of the larger community? Why or why not?” “How would you expect the following people to respond to the news that a halfway house is to be located in their neighbourhood: a parent of small children, a pastor, a senior? Why?” “Why might local residents be opposed to building a new subway station in their community?” “Who might oppose the redevelopment of local brownfields?”
- D1.2** investigate impacts on selected ecumenes of globalization and related trade policies and corporate activities
Sample questions: “How does the outsourcing of manufacturing and services from one country to another affect ecumenes in both countries?” “How can international trade policies and corporate actions influence disparities within or between ecumenes?” “How does globalization play a role in creating both ‘slums of hope’ and ‘slums of despair’?”

- D1.3** analyse the influence of past policies and historical attitudes on instances of conflict or cooperation in selected ecumenes

Sample questions: “How did issues relating to land claims and land use contribute to violent conflict in Caledonia, Ontario?” “How did the racial policies of the Belgian colonial administration in the 1930s contribute to the genocide in Rwanda in 1994?”

D2. Population Change and Migration

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

- D2.1** analyse impacts of population movements on cultural diversity and social interaction in urban areas
Sample questions: “How has the number of status Indians living in urban areas changed in comparison with the number living on reserves? What are some of the challenges that have arisen as a result of this change?” “How has recent migration from Asia altered Vancouver’s Chinatown?” “How do cultural communities encourage inclusion through events such as Mardi Gras in Sydney, Australia, Toronto’s Caribbean Carnival, and Kitchener-Waterloo’s Oktoberfest?”
Using spatial skills: Flow maps can be useful for identifying patterns in population

movements. Graphs of various kinds can help students identify and compare population patterns, changes, and trends.

D2.2 evaluate the capacity of a community's infrastructure and services (*e.g., transit, water, sanitation, housing, education, health care*) to accommodate population growth

Sample questions: "What criteria could be used to determine when a new school should be built? How do communities determine school locations?" "What is a reasonable amount of time to wait for public transit in different sizes of communities?" "Why do house prices vary so much from one ecumene to another?" "How should communities plan for population growth?"

D2.3 analyse impacts of different types of migration (*e.g., voluntary, forced, seasonal*) and other population changes (*e.g., growth, decline, aging*) on social conditions and the physical resources of cities and other ecumenes

Sample questions: "How do a city's services and facilities change in response to an aging population?" "Why might some companies choose not to invest in cities whose populations are getting older?" "How do cities respond to a decline in population?" "How do communities with economies based on new resource development cope with a rapid intake of population?" "How do communities in both source and destination regions benefit from seasonal migration? What related challenges do they face?"

Using spatial skills: Students can support their investigations of the impacts of migration on both source and destination regions by annotating a population base map with information about the ethnic backgrounds and other characteristics of incoming populations. Relevant information can be obtained from census data and immigration and emigration statistics.

D2.4 describe impacts of population diversity on countries that are major immigration destinations, and analyse policies for managing these impacts in selected countries (*e.g., Denmark, France, Japan, Australia, United States, Canada*)

Sample questions: "Why do large cities have ethnic neighbourhoods?" "How does diversity affect the liveability of a city?" "Which developed countries are the least diverse? Do they tend to have the most restrictive immigration policies?" "How do countries with more open immigration policies try to manage the impacts of diversity? What services do they provide to new immigrants? What protections do they offer?"

D2.5 analyse changes in urban growth in selected regions throughout the world, and assess impacts of these changes

Sample questions: "How have rates of urbanization differed over time and from one place to another?" "Why are rates of urbanization changing faster in some countries or regions than in others?" "How well are different cities coping with their current rate of change?" "How quickly is the rural/urban interface changing in a community near you?" "Should immigration policies be used to encourage urbanization? Why or why not?" "What impacts do uncontrolled migrations have on human settlements?"

Using spatial skills: Students can support their investigations of urbanization trends by adding text annotations and graphs to a base map of the region they have selected. Line graphs can indicate trends in population size over time. Circle graphs can show changes in population composition. Relevant data can be obtained from migration and city population statistics.

D3. Challenges of Interdependence

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

D3.1 assess the effectiveness of various mechanisms that countries use to assist each other (*e.g., development aid, food aid, debt relief, disaster relief*)

Sample questions: "Under what circumstances might developed countries receive aid from other countries?" "Should food aid be used to ensure long-term food security?" "Who benefits most from food aid?" "What is the most effective kind of aid for assisting victims of natural disasters?" "How can debt forgiveness help the development of an African nation?"

D3.2 analyse the impact of transnational criminal activities and informal economies (*e.g., black markets, grey markets*) on residents in developed and developing countries

Sample questions: "How does human trafficking connect the world?" "For what reason are humans trafficked?" "How are the following connected to informal economies: undocumented migrants, sweatshops, drug trade, prostitution?" "What challenges do governments have when working to shut down black-market activities?" "How does the relationship between those who are consumers of black-market products and those who are producers, or even products (*e.g., victims of human trafficking*), demonstrate an imbalance of power?"

D3.3 analyse impacts of consumerism in developed countries on ecumenes in developing countries (e.g., economic growth and job creation, increased resource consumption and pollution, migration to cities and the rise of informal settlements, greater potential for exploitation of labour as a result of demand for low-cost production)

Sample questions: “How are food supplies maintained when large numbers of people abandon farming in order to work in factories in cities?” “What responsibility do corporations in developing countries have to protect natural environments and workers’ rights in these countries?” “How does child labour affect communities in developing countries?” “What can consumers in Canada do to prevent the use of child labour or protect the rights of adult workers in foreign factories that produce goods

for their consumption?” “How do developing countries benefit from trade relations with developed countries? How can they build on those benefits while reducing the negative impacts that they have experienced?”

Using spatial skills: To support their investigations of links between consumerism and living conditions in developing countries, students can compare world thematic maps showing the number of computers per 100 inhabitants in each country and the amount of electronic waste exported and imported by each country. They can then add additional map layers showing socio-economic data, such as number of televisions per capita, number of doctors per 1,000 people, average years of schooling, and caloric intake to make inferences about linkages between consumption in developed countries and quality of life in developing countries.

E. CHANGING ECUMENES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1. Quality of Urban Life:** analyse interrelationships between urban environments and quality of life (**FOCUS ON:** *Spatial Significance; Interrelationships*)
- E2. Analysis of Solutions:** assess from a geographic perspective the effectiveness of solutions to issues affecting the built environment (**FOCUS ON:** *Interrelationships; Geographic Perspective*)

SPECIFIC EXPECTATIONS

E1. Quality of Urban Life

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

- E1.1** analyse factors affecting quality of life in urban environments generally and in selected communities (*e.g., water, sanitation, energy, and transportation infrastructure; educational, cultural, and recreational facilities; health care services; housing; employment; population density; security*)

Sample questions: “What influence does the built environment have on our quality of life?” “What services are most important for maintaining an acceptable quality of life? What services enhance the quality of life in urban communities?” “Why do people move from the countryside to cities? Why do people move from cities to suburbs?” “What are the negative effects of high population density? In what ways does high population density help to enhance quality of life? How does urban sprawl affect quality of life?” “How does the decay of a city core affect the quality of life of core residents?” “Does the selected community provide sufficient services for the population? Does it provide ample opportunity for employment? How would you improve the quality of life in this community?”

- E1.2** assess the role of planning (*e.g., land-use planning, infrastructure planning, green- and open-space planning, transportation flow planning*) in maintaining and enhancing the quality of life in urban communities

Sample questions: “Who is responsible for planning for quality of life?” “How can land-use decisions affect the quality of life of urban

residents?” “What strategies do planners use to protect and enhance the quality of life in a city?” “What factors need to be considered in a green-space plan if it is to improve the quality of life of everyone in a community?” “How do planners accommodate growth while limiting urban sprawl?” “What role does a city’s official plan play in its development?”

Using spatial skills: To support their analyses of the quality of life in built environments, students can use annotated maps of two urban communities of similar size to compare features that affect quality of life, such as schools, hospitals, transportation infrastructure, public green spaces, libraries and other cultural facilities, and sports and recreation facilities.

- E1.3** analyse factors affecting the quality of life in slums (*e.g., population density; types of building materials available; the state of infrastructure, including the type of water supply and the availability of facilities for waste and sewage treatment and disposal; access to medical care and education; the existence of class, racial, and other forms of discrimination*)

Sample questions: “Is quality of life necessarily tied to wealth?” “What is the difference between a ‘slum of hope’ and a ‘slum of despair’? What factors determine whether a slum will become a slum of hope or a slum of despair? “Could a slum of despair become a slum of hope? What could cause a slum of hope to become a slum of despair?” “How do surrounding areas respond to the presence of slums?” “What can be done to improve the quality of life in slums?” “Has globalization affected the quality of life in slums?”

E2. Analysis of Solutions

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

E2.1 analyse policy responses to social and economic challenges in selected ecumenes (e.g., crime, poverty, loss of an economic base, land redistribution, urban renewal)

Sample questions: “How did Elliott Lake respond to the loss of its major industry? How successful has it been in transforming its economic base?” “Why has South Africa renamed a number of communities, streets, and public buildings since the demise of apartheid? How have different groups in the country responded?” “Have farmland redistribution programs been successful in Zimbabwe?” “How is Argentina’s Social Housing Foundation attempting to improve living conditions for slum residents in Buenos Aires?” “Why would a community support a safe injection site?” “How can food banks improve the quality of life for all residents of the community?”

E2.2 assess, from a geographic perspective, the effectiveness of a selected megaproject as a solution to a local, regional, or national infrastructure need (e.g., the Three Gorges Dam in China; the James Bay Project in Quebec; the Aswan Dam in Egypt; the Big Dig in Boston; an Olympic site such as Beijing, Vancouver, London, or Sochi; the Mose flood barrier project in Venice)

Sample questions: “Are the benefits of the project worth the costs?” “What criteria can be used to determine the success of a megaproject?” “If the benefits of the project are enjoyed mostly outside of the ecumene in which it was built, can it still be considered a success?”